

SHAWN TRUJILLO JR

Ann Arbor, MI

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Education

University of Michigan

Master of Engineering in Electrical Computer Engineering, Embedded Systems

August 2026 – May 2027

Ann Arbor, Michigan

University of Michigan

Bachelor of Science in Computer Science, Minor in Electrical Engineering

August 2022 – May 2026

Ann Arbor, Michigan

Relevant Coursework

- Data Structures and Algorithms
- Computer Organization
- Computer Vision
- Computer Security
- Adv. Embedded Systems
- Logic Design
- Electronic Circuits
- Autonomous Robots
- Software Engineering
- Embedded Control
- Self-Driving Cars

Experience

Cleveland Cliffs

Summer '24, '25, '26

Process Control Intern | C, HTML/CSS, JavaScript, SQL

Dearborn, MI

- Developed a dynamic steel slab width optimization algorithm that identified compatible slab-width transitions across an entire daily casting schedule, reducing production waste and improving order fulfillment.
- Reduced monthly scrap by 60 tons, prevented 220 tons from being downgraded to secondary quality, and kept 1,100 tons on original prime orders through optimized scheduling.
- Created a background inventory tracking application that monitored steel plate locations throughout production, transfers, and shipping, providing real-time inventory visibility.
- Built six web applications that digitized safety audits and reporting for crane and railcar operators, supporting digital signatures, photo documentation, checklists, and offline data synchronization.
- Delivered presentations to plant leadership summarizing software projects, implementation strategies, and measurable operational benefits.

Undergraduate Recruitment - University of Michigan

January 2024 – Present

Senior Tour Guide

Ann Arbor, MI

- Led campus tours for groups of up to 30 prospective students and families, developing strong public speaking and communication skills.
- Mentored and trained new tour guides to ensure consistent tour quality and accurate university information.

Projects

Autonomous Robot Laser Tag | C++, Python, ROS2

March 2026

- Built a multi-robot IR communication system using ROS 2 and Raspberry Pi, enabling real-time hit detection.
- Designed a custom Raspberry Pi PCB integrating IR transmitters, receivers, and status LEDs for robot communication.
- Implemented a 38 kHz IR protocol with packet decoding and checksum validation with accuracy up to 8 feet.

Autonomous Guided Parafoil | C, Python

November 2025

- Designed and built a low-cost autonomous parafoil system to recover high-altitude payloads with precise, repeatable landings from heights above 50 meters.
- Developed STM32-based PCB Airborne Guidance Unit (FreeRTOS, custom HAL drivers) integrating GPS (u-blox NEO-M9N), BNO085 IMU, XBee radios, and servo-actuated brake lines.
- Implemented real-time sensor fusion using a custom Extended Kalman Filter (GPS + IMU) for accurate state estimation under dynamic flight conditions.
- Built a Python Ground Control Station for telemetry visualization, live tracking, and mission oversight via XBee radio link.

Autonomous Ballon Turret | C, Verilog, Python

March 2025

- Programmed MCU (STM32) to manage UART, SPI, and MMIO communication between camera, User-Interface, and motor control subsystems.
- Wrote custom VGA and LCD display drivers on FPGA for real-time system feedback.
- Integrated OpenCV on Raspberry Pi to detect balloon targets based on Hue-Saturation-Value and send coordinates to MCU based on color priority.
- Developed PWM-based control for servos and stepper motors to aim and fire accurately at balloon targets.

Technical Skills

Languages: Python, Java, C/C++/C#, HTML/CSS, JavaScript, SQL, Verilog

Developer Tools: VS Code, GitHub, XCode, STM32Cube, FreeRTOS, OpenCV, KiCad, ROS2